

CS302 – Modeling and Simulation

Course Placement

Modeling and Simulation is a core course offered to third year students of B.Tech Hons. ICT (Minor in Computational Science) program. It may be the foundation for domain based specialization electives such as Computational Finance, Computational and systems Biology and Computational Physics, Computational Social Science etc.

Course format

- 3 hours of lecture per week.
- 3 hours lab per week.

Prerequisites

Introductory Calculus, Probability and Statistics, Introductory Computational Physics and Numerical and computational methods. Some experience with scientific computing and programming (in Python) would be beneficial.

Course Content

This course introduces students to fundamentals of creating mathematical models of physical systems and implementation on computers to analyze the system both to gain insight and make predictions. The different mathematical approaches to modelling that are covered in the course can be characterized into differential and difference equation based models, probability based models which includes stochastic differential equations, cellular automata and event based approaches, and matrix based models. The course is interdisciplinary in nature and looks at systems from physics, biology, social science and engineering etc. from a modeling perspective. Each topic is followed by examples from different disciplines. The students are expected to test and create models of these systems in the lab, gaining intuitive understanding and report on it following the complete modelling approach.

Approach to be followed

system → Analyze the problem (determine problem's objective) → Formulate model (Gather data, make assumptions, determine equations/functions) → Solve model (Computer implementation) → Verify and interpret solutions → make predictions.

Text Books

- Angela B. Shiflet & George W. Shiflet. *Introduction to Computational Science: Modeling and Simulation for the Sciences*. Princeton University Press, 2006.
- Sheldon M. Ross, *Simulation*, Academic Press, 2006.
- Taylor & Karlin, *An Introduction to Stochastic Modeling*, Academic Press, 1998.
- S. Strogatz, *Nonlinear Dynamic and Chaos*, Oxford Universty Press, 2001.

Assessment method/Grading

Written exam: Exam: 60% (30+30)

Lab: 40% . The final lab will be on extensive simulation and will be treated as a project. **Any instance of copying or plagiarism will result in 0 for this component.**

Grading: Grading will be relative. Any student scoring less than 35% will receive a F grade.

Attendance: 80% attendance in both theory and lab is mandatory. Less attendance will result in grade reduction and/or not being allowed to participate in the exam.

Course Outcome

After successful completion of the course the students would be able to create a relevant model for a multitude of problems from science and engineering, by extracting the necessary and relevant information regarding the problem. They would also be able to define the different modeling terms by analyzing the system or the data that is present. They would be able to implement the model on the computer and from the results check for the validity of the model and correctness of the assumptions present in the model. They should be able to analyze the outcomes (mostly through visualizations) and make predictions. They would also be able to understand the limitations of their model and nuances in computer modeling of systems.

Course Content/Lecture schedule*

Sl. No.	Description
1	Overview of Computational Science
1.1	The Modeling Process.
1.2	Model Classification
1.3	Steps in the modeling Process.
2	System Dynamics Model (Compartment Models)
2.1	Unconstrained growth models.
2.2	Constrained Growth.
2.3	Applications: Radioactive decay, Drug dosage, population dynamics, growth of technologies, SI models for epidemic spread etc.
2.4	System of Equations:
2.5	Models from Epidemiology: epidemic spread models (SIR, SEIR, SEIZ etc.), predator-prey models, Modeling spread of infectious diseases such as SARS, Malaria
2.7	Models from Social Science: Rumor spread, Models for Innovation Diffusion etc.
2.8	Selective methods of analyzing nonlinear differential equations
3	Stochastic models (Monte Carlo simulations)
3.1	Review of basic probability
3.2	Generating random numbers from different distributions on computer.
3.3	Brownian motion and random walk in 1D, discrete and continuous time stochastic differential equation with applications.
3.4	Symmetric and asymmetric random walk, markovian and non markovian random walk
3.5	Markov Chains, Continuous Time Markov Chains, Birth and death processes, limiting time behavior.
4	Agent based Simulation (ABM)
4.1	Cellular Automata simulation of random walk
4.2	Ising Model for ferromagnetic Systems with applications to social science, metropolis algorithm
4.3	Applications: Modeling opinion formation on 2D lattice, segregation models, Rumor spread models etc.

* The number of topics covered will depend on the overall enthusiasm and commitment of the class.